Homework 3

2.2. ~~Then~~ the thermal average energy:

from Eqn. 47 chapter 1

$$U = -2sm\,B\,.$$

$$\Rightarrow \text{ spin excess} \rightarrow 2s = -\frac{U}{mB} \longrightarrow (1)$$

from Eqn. 35 chapter 1: we obtain the relation between multiplicity and spin excess:

$$g(N,s) \simeq g(N,0)\,\exp\left[-\frac{2s^2}{N}\right]$$

$$= g(N,0)\,\exp\left[-\frac{U^2}{2m^2B^2N}\right] \longrightarrow (2)$$

Hence the entropy:

$$\sigma = \log g(N,s)$$

$$\cong \log g(N,0) - \frac{U^2}{2m^2B^2N}$$

$$= \sigma_0 - \frac{U^2}{2m^2B^2N}$$

$$\Rightarrow \frac{1}{\tau} = \left.\frac{\partial\sigma}{\partial U}\right)_N = -\frac{U}{m^2B^2N} \longrightarrow (3)$$

Fractional Magnetization: $\dfrac{M}{Nm} = \dfrac{2\langle s\rangle}{N}$

from (1)

$$= -\frac{U}{mBN}$$

from (3)

$$= -\frac{U\,mB}{m^2B^2N} = \frac{mB}{\tau}$$

$$\Rightarrow \boxed{\frac{M}{Nm} = \frac{mB}{\tau}}$$

2.3. (a) From Eqn (1.55)

multiplicity of a set of N oscillators:

$$g(N,n) = \frac{(N+n-1)!}{n!\,(N-1)!}$$

$$\sigma = \log g(N,n)$$

$$= \log(N+n-1)! - \log n! - \log(N-1)!$$

Using the Stirling approximation:

$$\log N! \simeq N\log N - N$$

$$\Rightarrow \log(N+n-1)! \simeq (N+n-1)\log(N+n-1) - (N+n-1)$$

$$\simeq (N+n)\log(N+n) - (N+n)$$

$$\Rightarrow \log n! \simeq n\log n - n$$

$$\Rightarrow \log(N-1)! \simeq (N-1)\log(N-1) - (N-1)$$

$$\simeq N\log N - N$$

(b)

$$\sigma \simeq (N+n)\log(N+n) - (N+n)$$

$$- (n\log n - n)$$

$$- (N\log N - N)$$

$$\Rightarrow \sigma \simeq (N+n)\log(N+n) - n\log n - N\log N$$

Use $\rightarrow$ $U = n\hbar\omega$

$$\Rightarrow \sigma \simeq \left[N + U/\hbar\omega\right]\log\left(N + \frac{U}{\hbar\omega}\right) - \left(\frac{U}{\hbar\omega}\right)\log\left(\frac{U}{\hbar\omega}\right) - N\log N$$

$$\frac{1}{\tau} = \frac{\partial \sigma}{\partial u}$$

$$= \frac{1}{\hbar\omega} \log\left(N + \frac{u}{\hbar\omega}\right) - \frac{1}{\hbar\omega}$$

$$- \frac{1}{\hbar\omega} \log\left(\frac{u}{\hbar\omega}\right) + \frac{1}{\hbar\omega}$$

$$\Rightarrow \quad \frac{\hbar\omega}{\tau} = \log\left[\frac{N\hbar\omega + u}{u}\right]$$

$$\exp\left[\frac{\hbar\omega}{\tau}\right] = \frac{N\hbar\omega}{u} + 1$$

$$u = \frac{N\hbar\omega}{\exp\left[\frac{\hbar\omega}{\tau}\right] - 1}.$$

P4

$\boxed{2.5}$

(a) $$\frac{g_1(N_1, \hat{S}_1+\delta)\, g_2(N_2, \hat{S}_2-\delta)}{(g_1 g_2)_{max}} = \exp\left[-\frac{2\delta^2}{N_1} - \frac{2\delta^2}{N_2}\right]$$

$$N_1 \simeq N_2 = 10^{22}$$

$S_1 = \hat{S}_1 + 10^{11} \Rightarrow \hat{\delta} = 10^{11}$

$$\Rightarrow \frac{g_1 g_2}{(g_1 g_2)_{max}} = \exp\left[-\frac{2\,10^{22}}{10^{22}} - \frac{2\,10^{22}}{10^{22}}\right]$$

$$= \exp[-4] \simeq 0.0183$$

(b) $S' = 10^{20}$, $N_1 = N_2 = 10^{22}$

$$(g_1 g_2)_{max} = g_1(0)\, g_2(0) \exp\left[-\frac{2S'^2}{N}\right] \longrightarrow (15)$$ ← CHAPTER(2)

From Eqn. (13)

$$\frac{S_1}{N_1} = \frac{S - S_1}{N_2} \Rightarrow S_1 = S - S_1 \quad [\text{since } N_1 = N_2]$$

$\Rightarrow S_1 = S/2$ ← For most probable configuration only:

From Eqn. (14) →

$$\frac{S_1}{N_1} = \frac{S}{N} \Rightarrow N = \frac{S}{S_1} N_1 = 2 \times 10^{22}$$

i.e $N = N_1 + N_2$

$$(g_1 g_2)_{max} = g_1(0)\, g_2(0) \exp\left[-2\,\frac{\times 10^{40}}{2\times 10^{22}}\right]$$

$$= g_1(0)\, g_2(0) \exp\left[-10^{18}\right]$$

$$\sum_{s_1=0}^{10^{22}} g_1(N_1, s_1)\, g_2(N_2, s-s_1) = g_1(0)\, g_2(0) \underbrace{\sum_{s_1=0}^{10^{22}} \exp\left[\frac{-2s_1^2}{N_1} - \frac{2(s-s_1)^2}{N_2}\right]}_{Sum}$$

$$N_1 = N_2$$

$$Sum = \sum_{s_1=0}^{10^{22}} \exp\left[-\frac{2s_1^2}{N_1} - \frac{2(s^2+s_1^2-2ss_1)}{N_1}\right]$$

$$= \sum_{s_1=0}^{10^{22}} \exp\left[-\frac{(4s_1^2 - 4ss_1 + s^2)}{N_1} - \frac{s^2}{N_1}\right]$$

$$= \exp\left[-\frac{s^2}{N_1}\right] \sum_{s_1=0}^{10^{22}} \exp\left[-\frac{(2s_1 - s)^2}{N_1}\right]$$

this summation can be approximate as an integral

$$\approx \exp\left[-\frac{s^2}{N_1}\right] \int_0^{\infty} ds_1 \exp\left[-\frac{(2s_1 - s)^2}{N}\right]$$

Choose $k = 2s_1 - s \Rightarrow dk = 2\, ds_1$

Note when $s_1 = 0 \Rightarrow k = -s = -10^{22} \longrightarrow -\infty$

$$= \exp\left[-\frac{s^2}{N_1}\right] \int_{-\infty}^{\infty} \left(\frac{dk}{2}\right) \exp\left[-\frac{k^2}{N}\right]; \quad \bar{k} = \frac{k}{\sqrt{N}}$$

$$= \exp\left[-\frac{s^2}{N_1}\right] \frac{\sqrt{N}}{2} \int_{-\infty}^{\infty} d\bar{k} \exp\left[-\bar{k}^2\right] = \frac{\sqrt{N\pi}}{2} \exp\left[-\frac{s^2}{N_1}\right]$$

thus $\rightarrow$

$$\sum_{s_1=0}^{10^{22}} g_1(N_1, s_1)\, g_2(N_2, s-s_1) \approx g_1(0)\, g_2(0)\, \frac{\sqrt{N\pi}}{2} \exp\left[-\frac{s^2}{N_1}\right]$$

$$(g_1 g_2)_{max} = g_1(0)\, g_2(0) \exp\left[-\frac{2s^2}{N}\right] = g_1(0)\, g_2(0) \exp\left[-\frac{s^2}{N_1}\right]$$

$$\Rightarrow \frac{\sum_{s_1=0}^{10^{22}} g_1(N_1, s_1)\, g_2(N_2, s-s_1)}{(g_1 g_2)_{max}} = \frac{\sqrt{N\pi}}{2} = \sqrt{N_1\pi/2}$$

$$\Delta\sigma = \ln \sum_{s_1=0}^{10^{22}} g_1 g_2 - \ln (g_1 g_2)_{max} = \ln\left[\frac{\sum_{s_1=0}^{10^{22}} g_1 g_2}{(g_1 g_2)_{max}}\right]$$

$$= \ln\sqrt{N_1\pi/2} \approx \frac{1}{2}\log N_1 \approx 25.3$$

$$\hat{s}_1 = \hat{s}_2 = \frac{s}{2}\ ;\ (g_1 g_2)_{max} = \left[\frac{2}{\pi N_1}\right] 2^{2N_1} \exp\left[-\frac{2s_1^2}{N_1}\right] \exp\left[-\frac{2s_2^2}{N_2}\right]$$

$N_1 = N_2 \Rightarrow$

$$= \left[\frac{2}{\pi N_1}\right] 2^{2N_1} \exp\left[-\frac{4s_1^2}{N_1}\right]$$

$$= \left[\frac{2}{\pi N_1}\right] 2^{2N_1} \exp\left[-\frac{s^2}{N_1}\right]$$

$$\sigma_{max} = \log (g_1 g_2)_{max} = \ln\left[\frac{2}{\pi N_1} 2^{2N_1}\right] - \frac{s^2}{N_1}$$

$$= \ln\frac{2}{\pi} + 2N_1 \log 2 - 22\log 10 - \frac{s^2}{N_1}$$

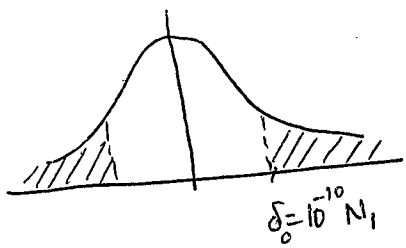
$$\approx 2N_1 \log 2 \approx 1.38 \times 10^{22}.$$

$$\Rightarrow \quad \frac{\Delta\sigma}{\log[g_1 g_2]_{max}} \approx 18.3 \times 10^{-22} \leftarrow \text{very small}$$

(6)

$$\frac{g_1(N_1, \hat{S}_1 + \delta)\, g_2(N_2, \hat{S}_2 - \delta)}{(g_1 g_2)_{max}} = \exp\left[-\frac{4\delta^2}{N_1}\right]$$

$$\frac{\delta}{N_1} > 10^{-10} \Rightarrow \delta > 10^{-10} N_1 \; ; \quad N_1 = 10^{22}$$



We have to calculate the shaded area and divide it by the total area to get the fraction.

$$\text{Shaded area} \equiv 2 \int_{\delta_0}^{\infty} \exp\left[-\frac{4\delta^2}{N_1}\right] d\delta \; ; \quad \frac{2\delta}{\sqrt{N_1}} = k$$

$$= \sqrt{N_1} \int_{\frac{2\delta_0}{\sqrt{N_1}}}^{\infty} \exp[-k^2]\, dk$$

$$\text{Total Area} = \sqrt{N_1} \int_0^{\infty} \exp[-k^2]\, dk$$

$$\text{Ratio} = \frac{\int_{2\delta_0/\sqrt{N_1}}^{\infty} \exp[-k^2]\, dk}{\int_0^{\infty} \exp[-k^2]\, dk} = \frac{\int_{2\times 10^{-10}\sqrt{N_1}}^{\infty} \exp(-k^2)\, dk}{(\sqrt{\pi}/2)}$$

$$\text{Ratio} = \frac{2}{\sqrt{\pi}} \int_{z_0}^{\infty} \exp(-k^2)\, dk \approx 5 \times 10^{-176}$$